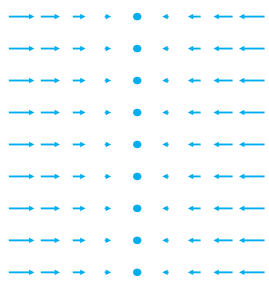
Suggested problems: 1-29 odd

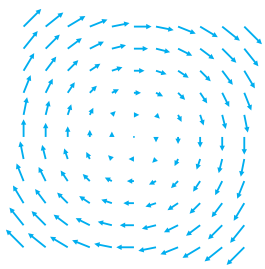
Problem 1: For parts (a)-(f), does the vector field appear to be conservative?

(a)



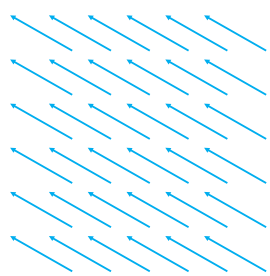
Yes

(b)



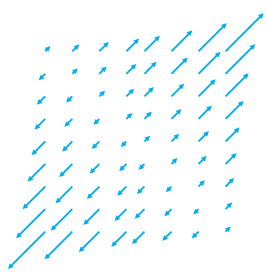
No

(c)



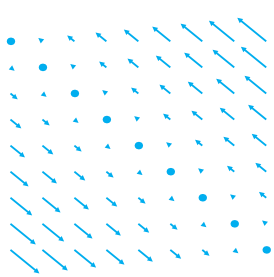
Yes

(d)



Yes

(e)

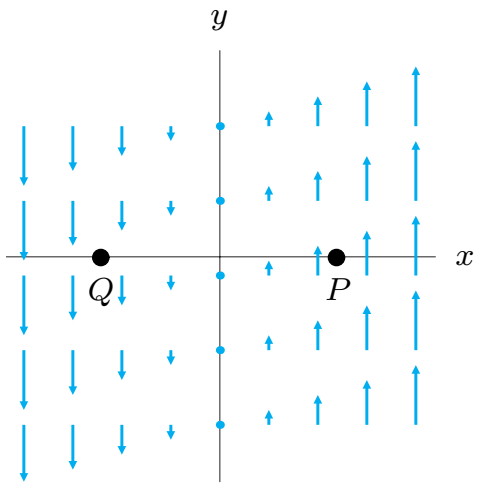


No

Problem 2: The figure below shows the vector field $\vec{F} = \langle 0, x \rangle$. Find paths C_1, C_2, and C_3 from P to Q such that

$$\int_{C_1} \vec{F} \cdot d\vec{r} = 0, \quad \int_{C_2} \vec{F} \cdot d\vec{r} > 0, \quad \int_{C_3} \vec{F} \cdot d\vec{r} < 0.$$

Is $\vec{F}$ a gradient field?



See class notes.

Problem 3: Prove that $\vec{F}(x, y) = \langle 2xy, xy \rangle$ is not conservative.

The field is not conservative because it has nonzero curl:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = y - 2x \neq 0.$$

Problem 4: Show that the field

$$\vec{F}(x, y) = \frac{\langle -y, x \rangle}{x^2 + y^2}$$

has zero curl on its domain (the plane with the origin removed), but is *not* conservative.

The curl is

$$\begin{aligned} \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} &= \frac{(x^2 + y^2) - 2x^2}{(x^2 + y^2)^2} - \frac{2y^2 - (x^2 + y^2)}{(x^2 + y^2)^2} \\ &= \frac{y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - x^2}{(x^2 + y^2)^2} \\ &= 0. \end{aligned}$$

To show that $\vec{F}$ is not conservative, we will find a closed path C over which $\oint \vec{F} \cdot d\vec{r}$ is not zero. The curve we will use is

$$\vec{r}(t) = \langle \cos(t), \sin(t) \rangle, \quad t \in [0, 2\pi].$$

Then

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} dt = 2\pi.$$

Problem 5: Decide which of the following vector fields are conservative. For the ones that are conservative, find a potential function.

(a) $\vec{F}(x,y) = \langle y, -x \rangle$

We check the curl:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -2.$$

Since this is nonzero, $\vec{F}$ is not conservative.

(b) $\vec{F}(x,y) = \langle 2xy, x^2 \rangle$

We check the curl:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

Since the domain of $\vec{F}$ is simply connected, the Curl Test says that $\vec{F}$ is conservative. Then

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2xy \quad \Rightarrow \quad f(x,y) = x^2y + C_1(y), \\ \frac{\partial f}{\partial y} &= x^2 \quad \Rightarrow \quad f(x,y) = x^2y + C_2(x), \end{aligned}$$

so $f(x,y) = x^2y$.

(c) $\vec{F}(x,y) = \langle 2xy^3 + y, 3x^2y^2 + x \rangle$

We check the curl:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

Since the domain of $\vec{F}$ is simply connected, the Curl Test says that $\vec{F}$ is conservative. Then

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2xy^3 + y \quad \Rightarrow \quad f(x,y) = x^2y^3 + xy + C_1(y), \\ \frac{\partial f}{\partial y} &= 3x^2y^2 + x \quad \Rightarrow \quad f(x,y) = x^2y^3 + xy + C_2(x), \end{aligned}$$

so $f(x,y) = x^2y^3 + xy$.

(d) $\vec{F}(x,y) = \dfrac{\langle x, y \rangle}{x^2 + y^2}$

We check the curl:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

But since the domain of $\vec{F}$ is *not* simply connected, the Curl Test does not apply. Hence $\vec{F}$ may be conservative, or may not be. So, we attempt to find a potential function anyways, not knowing whether or not it even exists. Note that:

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{x}{x^2+y^2} \quad \Rightarrow \quad f(x,y) = \frac{1}{2}\ln\left(x^2+y^2\right) + C_1(y), \\ \frac{\partial f}{\partial y} &= \frac{y}{x^2+y^2} \quad \Rightarrow \quad f(x,y) = \frac{1}{2}\ln\left(x^2+y^2\right) + C_2(x), \end{aligned}$$

so $f(x,y) = \frac{1}{2}\ln\left(x^2+y^2\right)$. One can check that $\nabla f = \vec{F}$, so $\vec{F}$ is indeed conservative.